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VOL. 01 · EDITION I

THE LLM BUILDER'S SERIES

Finetuning *Dictionary.*

26 essential LLM finetuning terms — explained simply.



Chandra Sekhar
AI Engineer & Trainer

INSIDE

26 essential terms

READ TIME

3 minutes

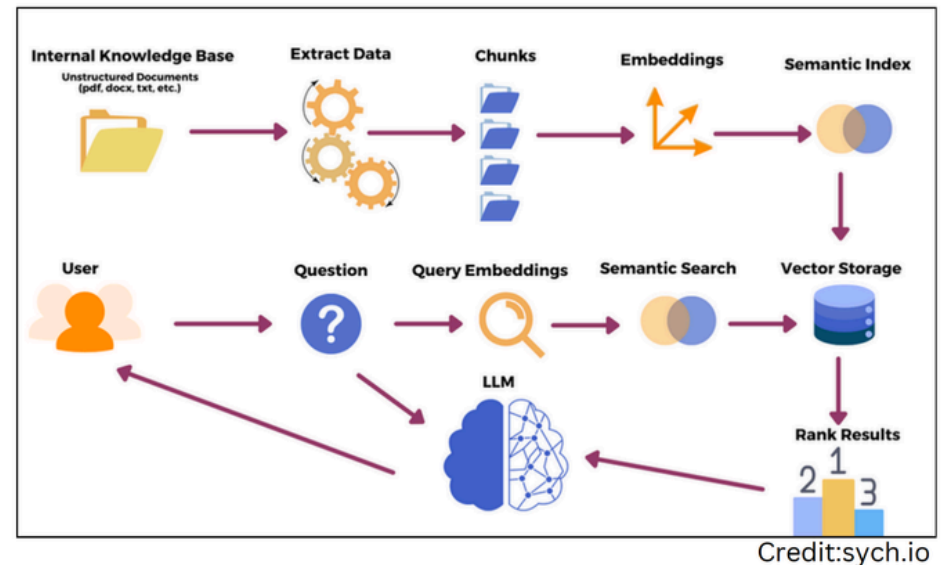
PAGES

A — Z

A

Augmentation

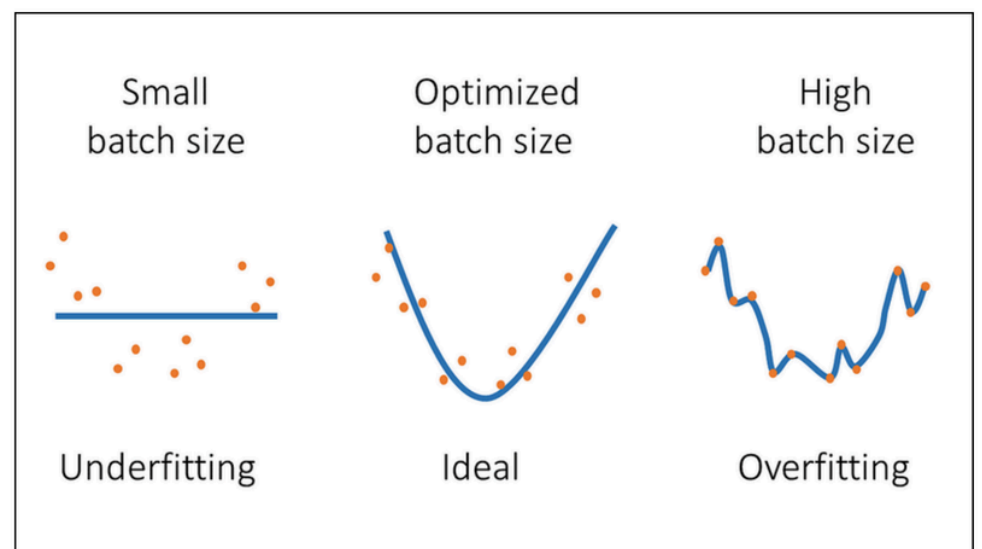
- Expanding model abilities with external data or techniques.
- Ensures better outputs by adding domain knowledge.
- Enhances reliability in fine-tuned LLMs.
- Common in adaptive learning systems.



B

Batch Size

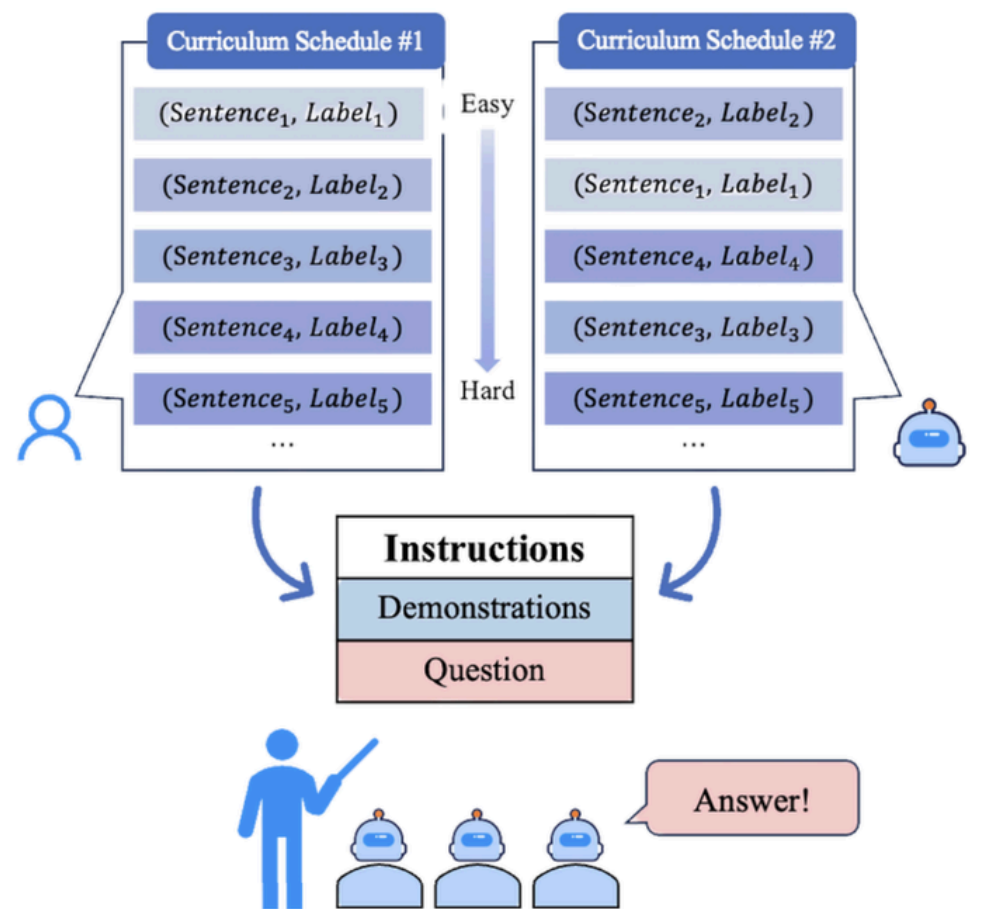
- The number of samples processed before updating model weights.
- Affects learning stability and speed.
- Balancing small and large batch sizes is key.
- Important for tuning computational efficiency.



C

Curriculum Learning

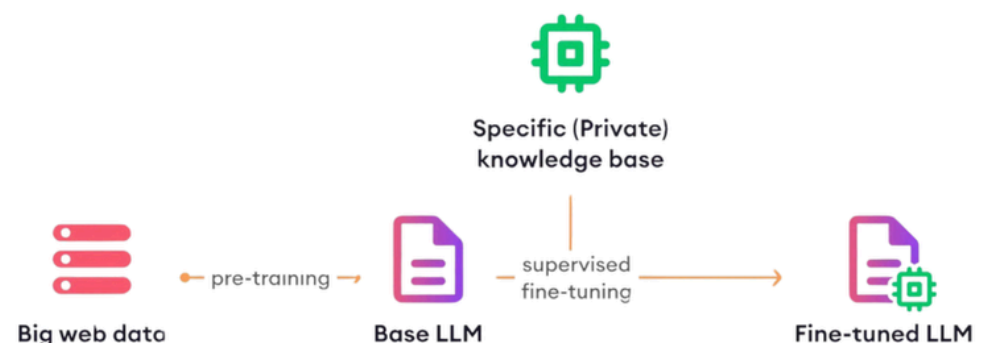
- Training models step-by-step with increasing complexity.
- Speeds up convergence.
- Helps the model generalize better to tasks.
- Inspired by human learning systems.



D

Domain-Specific Tuning

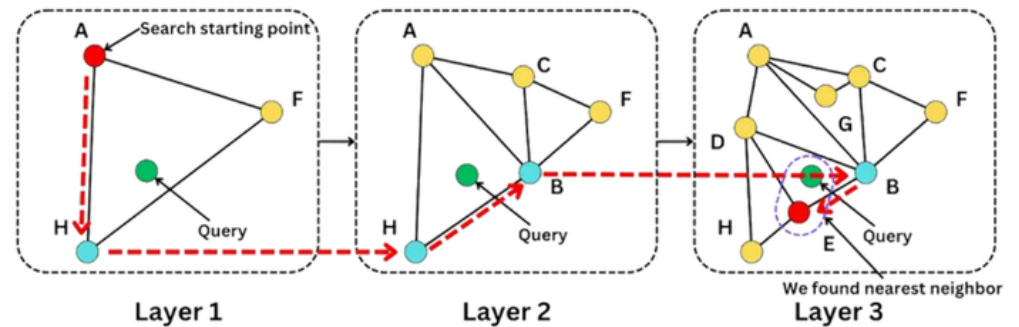
- Adapting LLMs to excel in specialized fields.
- Essential for medical, legal, or finance sectors.
- Improves performance in niche applications.
- Requires high-quality labeled datasets.



E

Embeddings

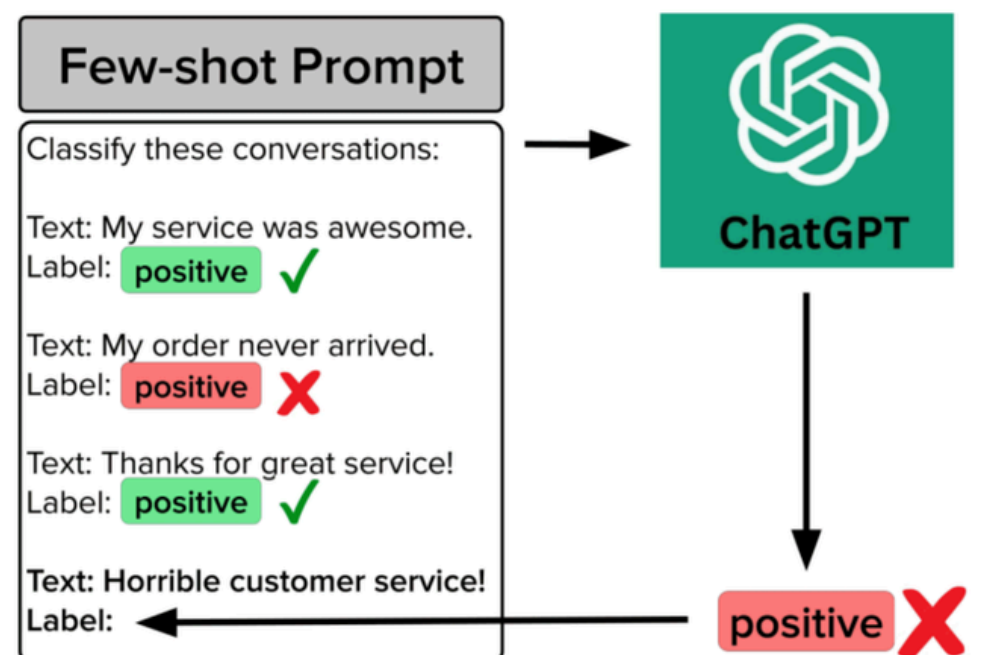
- Converting text into numerical vectors for analysis.
- Key for semantic search & dense retrieval.
- Used extensively in fine-tuning pipelines.
- Powers tasks like recommendations and clustering.



F

Few-Shot Learning

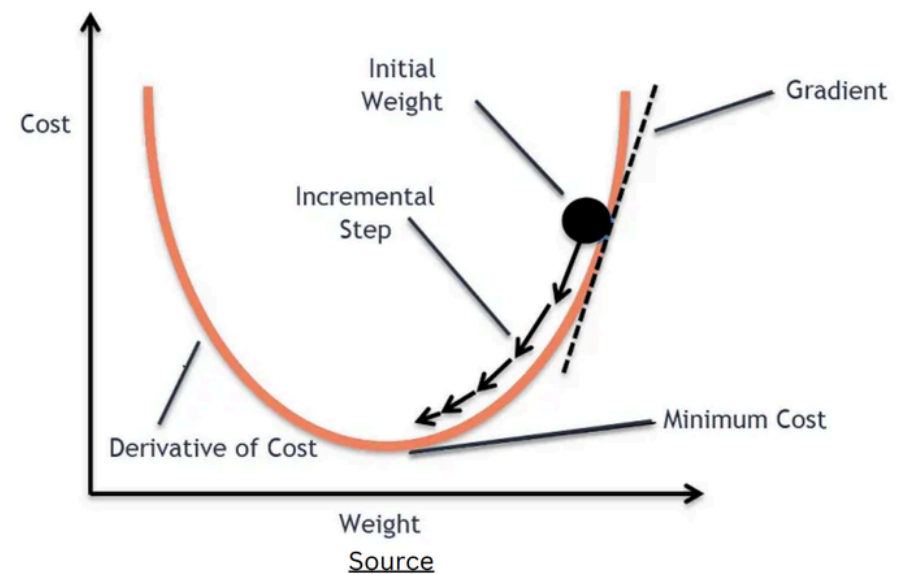
- Fine-tuning with minimal labeled data.
- Enables rapid adaptation to new tasks.
- Helps in low-data scenarios.
- Balances pre-training and specialized knowledge.



G

Gradient Descent

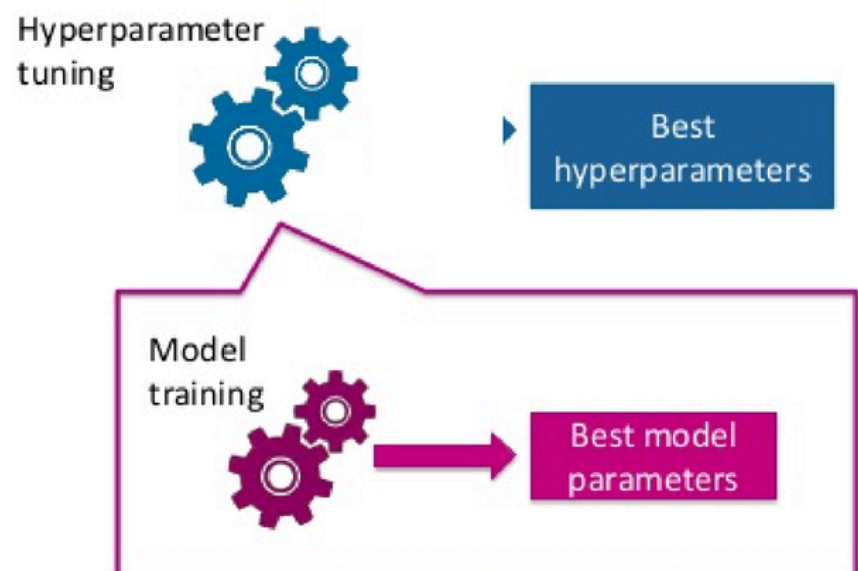
- Optimization method to minimize error in training.
- Drives the learning process by adjusting weights.
- Central to every fine-tuning process.
- Requires proper tuning of learning rates.



H

Hyperparameters

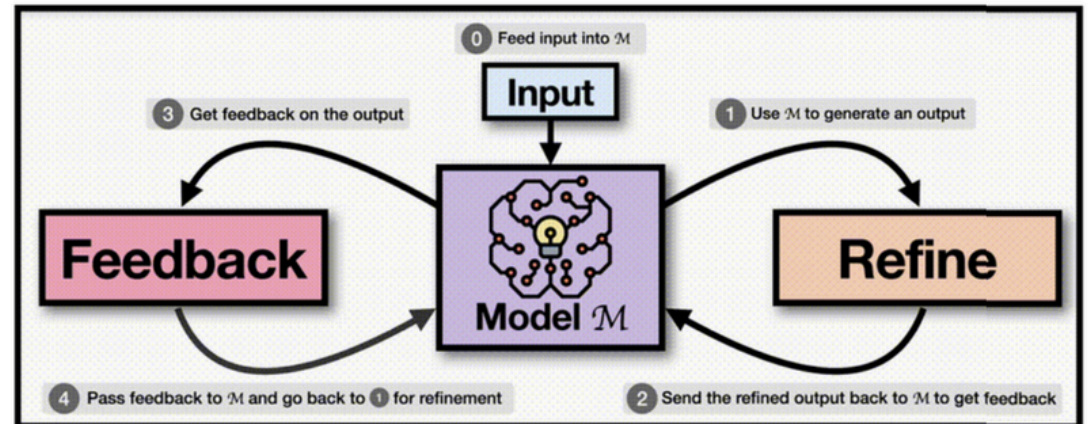
- Settings like batch size, learning rate, and epochs.
- Fine-tuning requires precise hyperparameter optimization.
- Impacts model accuracy and speed.
- Experimentation helps achieve optimal results.



I

Iterative Training

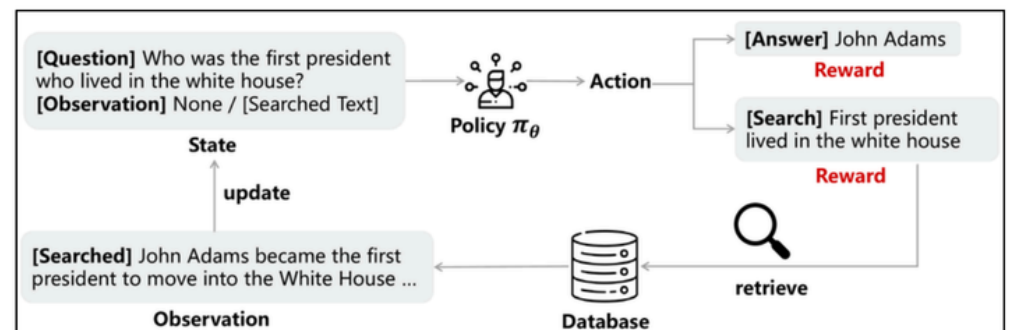
- Refining models through repeated training cycles.
- Enhances performance step by step.
- Key for achieving state-of-the-art results.
- Reduces overfitting by monitoring progress.



J

Joint Learning

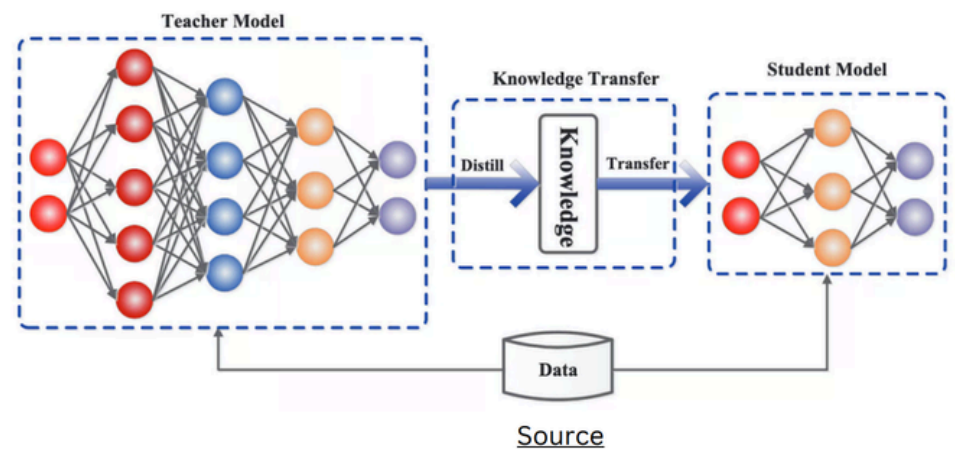
- Trains retrieval and generation components simultaneously.
- Ensures better synergy between retrieved data and generated outputs.
- Reduces the need for separate fine-tuning.
- Often leads to improved overall system performance.



K

Knowledge Distillation

- Transferring knowledge from large models to smaller ones.
- Makes models more efficient for deployment.
- Retains essential capabilities with fewer resources.
- Common in low-resource environments.

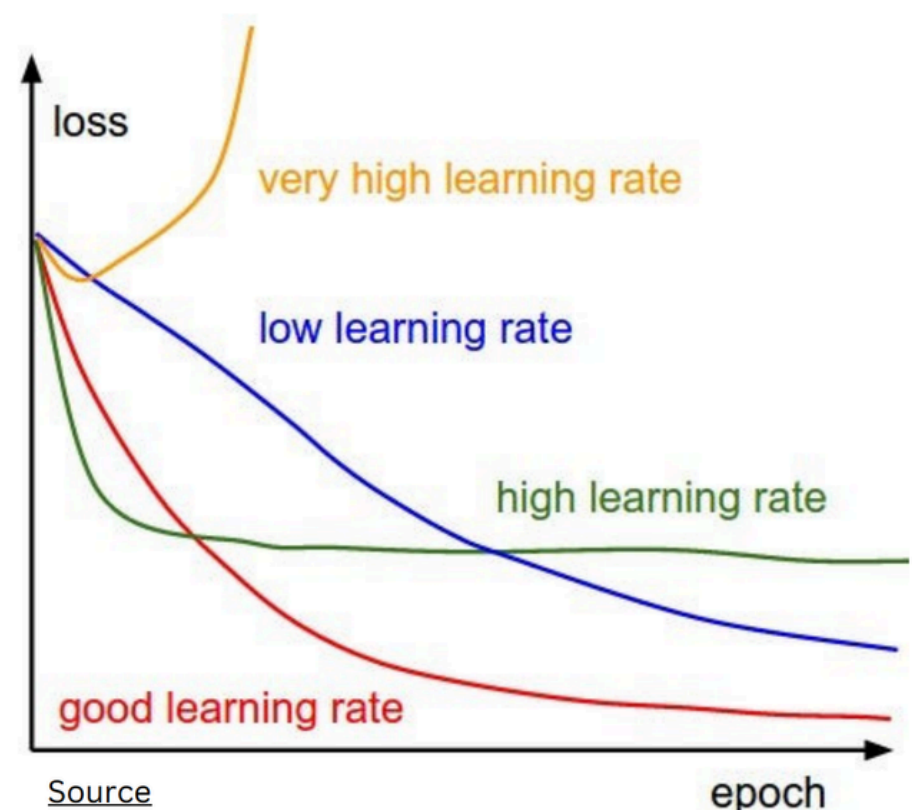


L

Learning Rate

- The speed at which the model learns.
- Critical to balance between slow and fast learning.
- Improper tuning can lead to overfitting or underperformance.
- Often adjusted dynamically during training.

L - Learning Rate



M

Model Weights

- Parameters learned during training.
- Define how the model processes input.
- Fine-tuning adjusts weights for specific tasks.
- Pretrained weights act as a foundation.

Finetuned Weights

Weight Update

$$\overbrace{W_{ft}}^{\text{Finetuned Weights}} = \underbrace{W_{pt}}_{\text{Pretrained Weights}} + \overbrace{\Delta W}^{\text{Weight Update}}$$

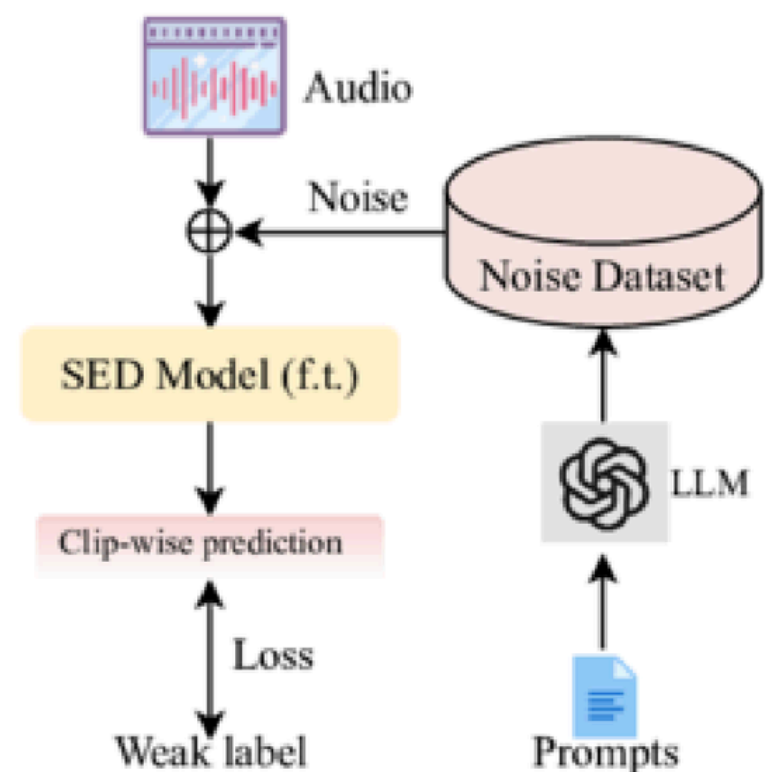
Pretrained Weights

Source

N

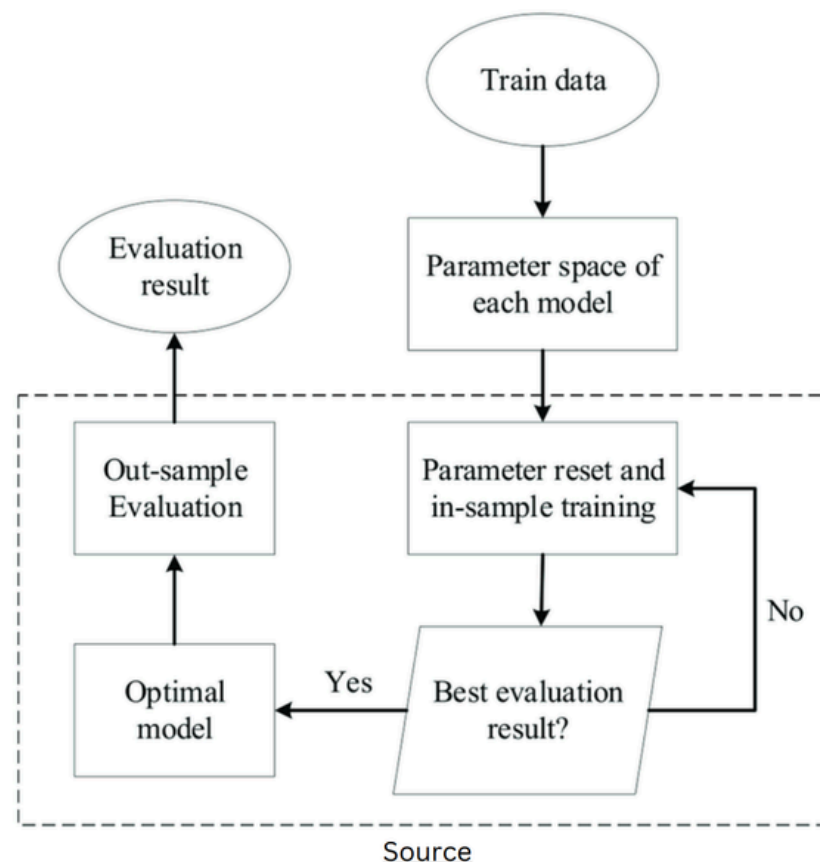
Noise Handling

- Addressing noisy or low-quality data in fine-tuning.
- Improves the model's ability to generalize.
- Requires robust preprocessing pipelines.
- Common in messy, real-world datasets.



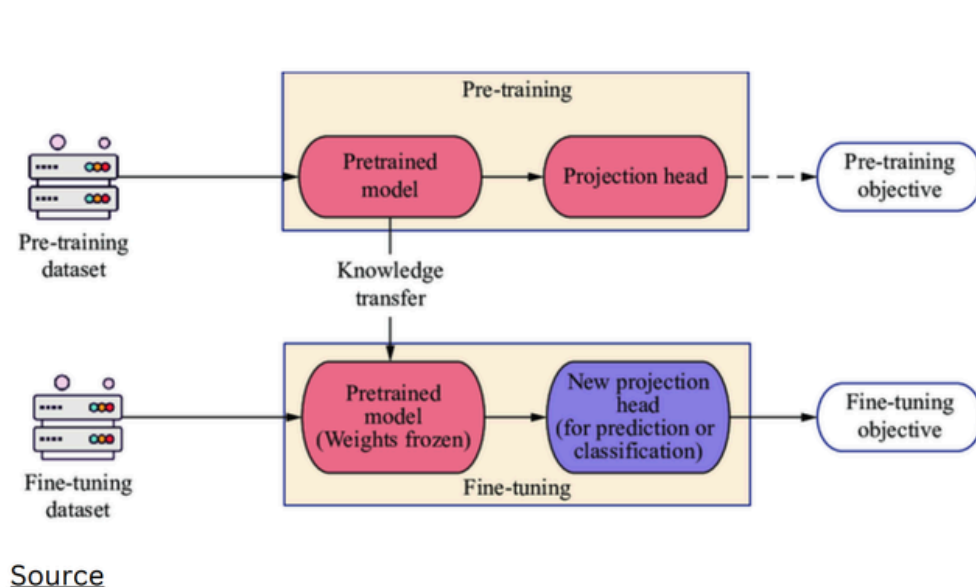
Optimization

- The process of improving model performance.
- Involves methods like Adam or SGD.
- Essential for faster convergence in fine-tuning.
- Balances trade-offs between accuracy and efficiency.



Pretraining

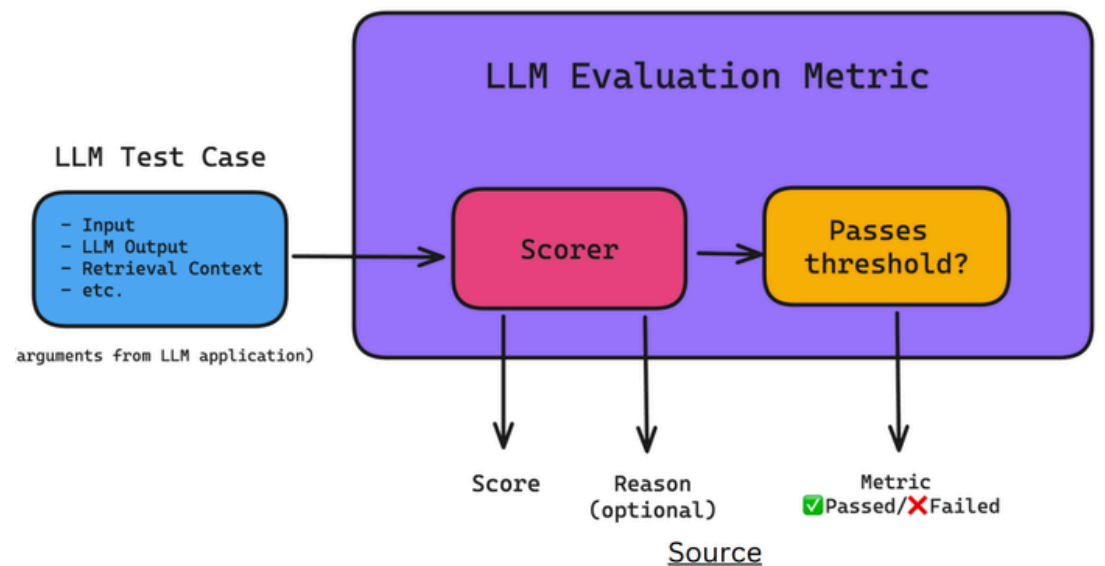
- Training on large datasets before fine-tuning.
- Provides general knowledge to the model.
- Reduces data requirements for fine-tuning.
- Speeds up task-specific adaptation.





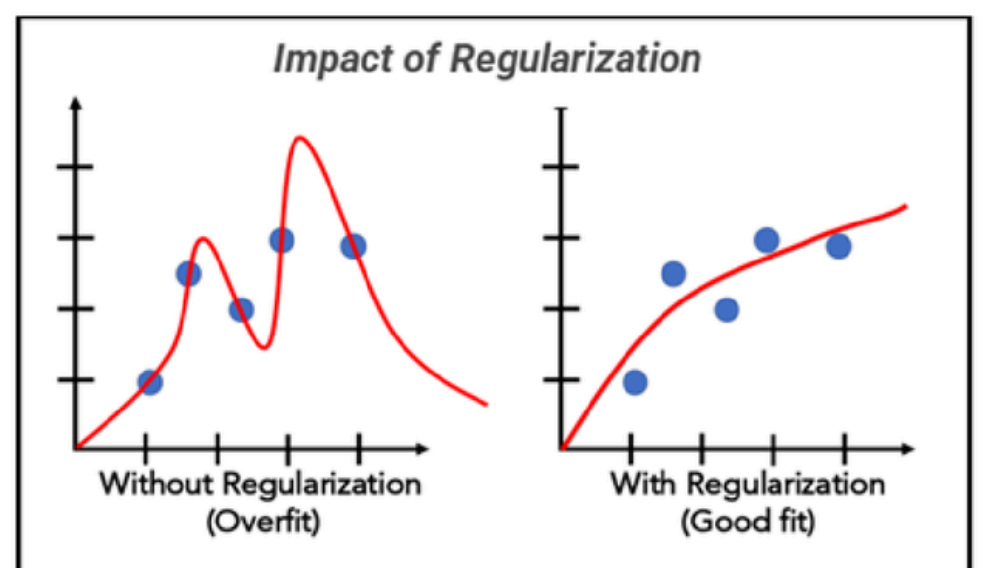
Quality Evaluation

- Measuring performance with metrics like BLEU, ROUGE, or accuracy.
- Validates the effectiveness of fine-tuning.
- Guides iterative improvements.
- Helps identify issues like overfitting.



Regularization

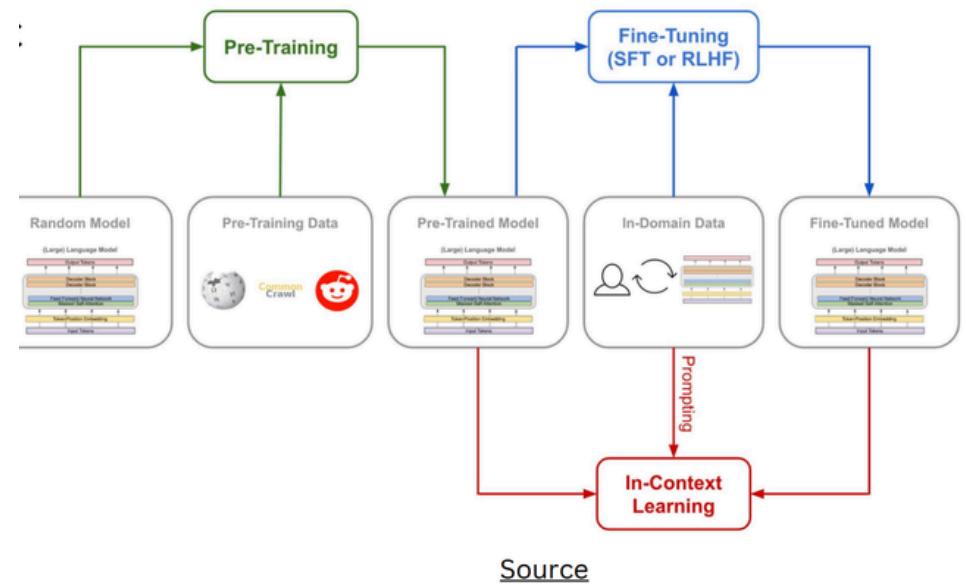
- Techniques to prevent overfitting.
- Includes dropout, weight decay, or early stopping.
- Improves generalization on unseen data.
- Key for robust fine-tuned models.



S

Supervised Fine-Tuning

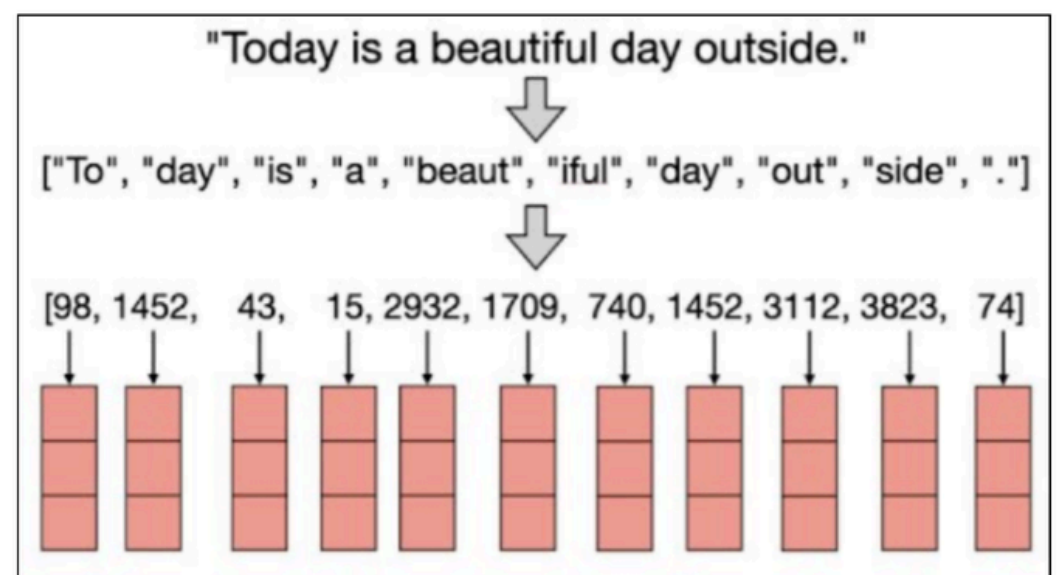
- Using labeled data to teach specific tasks.
- Boosts performance in well-defined use cases.
- Requires high-quality annotations.
- Common in domain-specific applications.



T

Tokenization

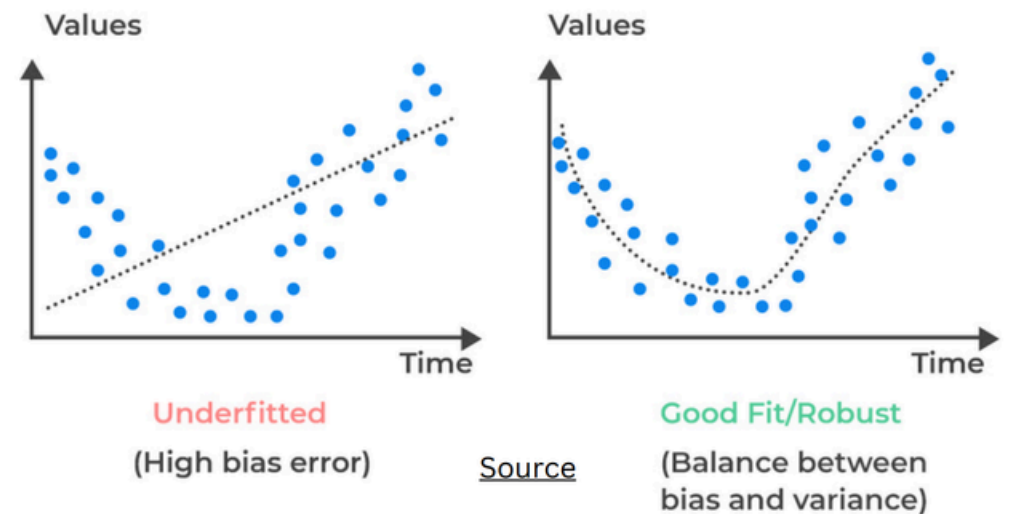
- Splitting text into smaller units (tokens).
- Prepares data for model consumption.
- Handles variations like punctuation and casing.
- Fundamental for both training and inference.



U

Underfitting

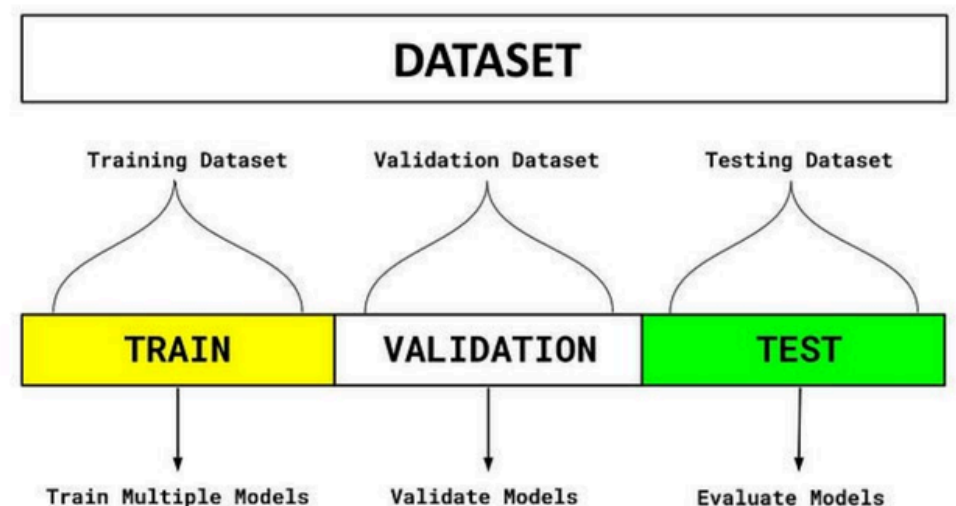
- A model failing to capture data patterns.
- Often due to insufficient training.
- Addressed by increasing complexity or data size.
- Opposite of overfitting.



V

Validation Set

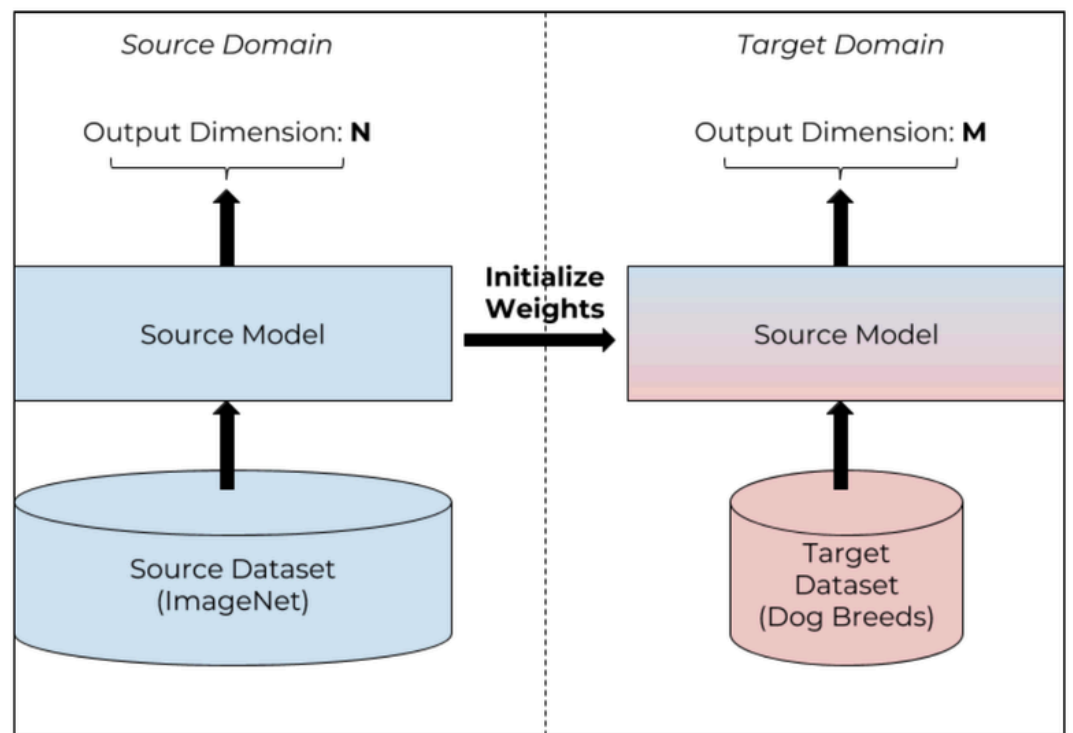
- Dataset used to monitor training performance.
- Helps prevent overfitting during fine-tuning.
- Guides decisions on hyperparameter tuning.
- Ensures the model generalizes well.



W

Warm-Start

- Initializes retrieval systems with pre-trained embeddings or models.
- Speeds up convergence and improves early-stage performance.
- Reduces training time for new tasks.
- Common in transfer learning scenarios.

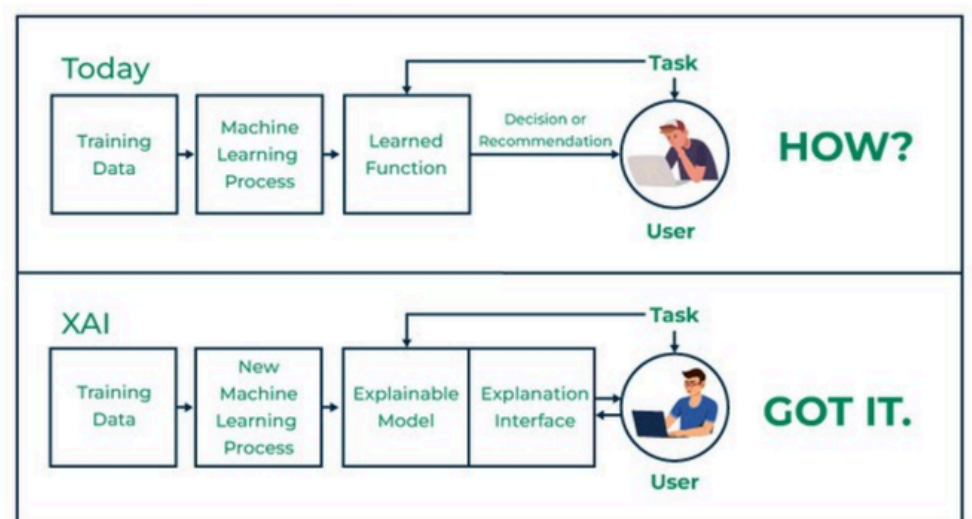


Credit:determined.ai

X

Explainability

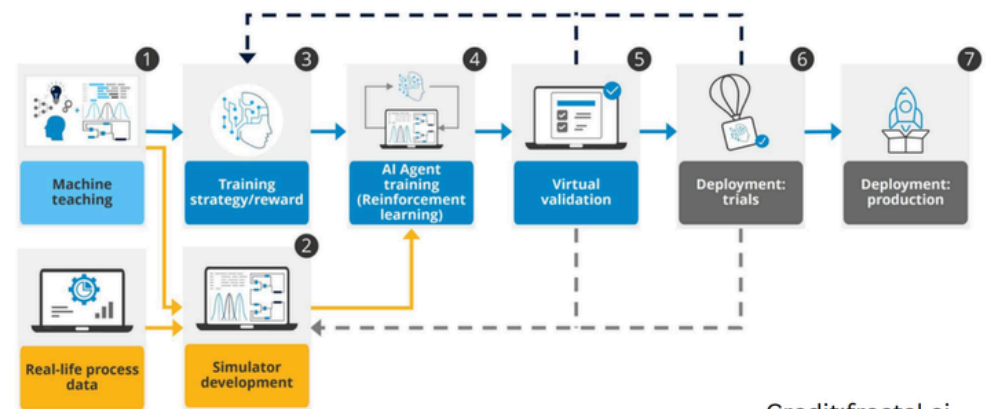
- Understanding model predictions.
- Essential for building trust in AI systems.
- Identifies biases or issues in fine-tuned outputs.
- Key for high-stakes applications like healthcare.



Y

Yield Optimization

- Maximizing output relevance and efficiency.
- Improves response quality for fine-tuned models.
- Involves iterative adjustments and monitoring.
- Enhances user satisfaction in real-world use.

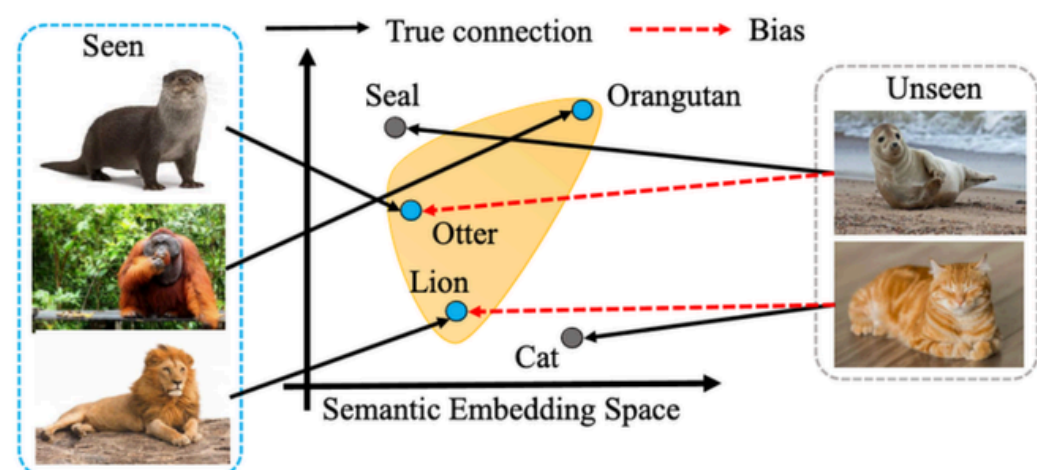


Credit:fractal.ai

Z

Zero-Shot Learning

- Performing tasks without task-specific fine-tuning.
- Leverages general knowledge from pretraining.
- Useful for quick adaptation to new domains.
- A hallmark of advanced LLMs.



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Chandra Sekhar

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